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Forced oscillation suppression using extended virtual synchronous generator in a low-Inertia microgrid

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ABSTRACT

Effects of forced oscillations (FOs) in well-damped power systems are relatively smaller than those of microgrids (MGs) in which the severity of the FOs may be intensified by converter interfaced generators (CIGs). According to the distinct system characteristics, the FOs in MGs will be challenging problems in future research topics. Without proper controls of the CIGs, the FOs may be exhibited extremely higher amplitude, resulting in the MG instability. Such influences will be exacerbated in a low-inertia MG, which can trigger critical frequency oscillation and system collapse. This paper introduces an extended virtual synchronous generator (VSG) with virtual forced components to attenuate the dynamic FO effects in the presence of a low-inertia MG. Contrastive scenarios, i.e., periodic, combined, full sine, and high-frequency FOs are conducted to validate the performances of VSG control in both stand-alone and interconnected MG environments. Numerical results verify that the extended VSG control provides promising benefits in a low-inertia MG not only for the sake of MG stability improvement but also the further FO suppression.

1. Introduction

Environmental problems such as global warming, air pollution, and fossil fuel depletion raise critical concerns for traditional resources, leading to concentrations on zero-emission energy resources, e.g., renewable energy sources (RESs). Accordingly, the infrastructure of traditional power systems is changing to be more energy independent and more efficiently known as microgrids (MGs). The MGs locally represent distributed energy grids (i.e., small-scale power systems) that are efficient and optimal for integrating RES-based converter interfaced generators (CIGs), such as wind-turbine generators (WTGs) and solar/photovoltaic (PV) systems. Subsequently, CIGs have become a main component in the existing MGs. With the increasing CIGs in MGs, the CIGs introduce a new stability issue, which can trigger the lack of physical inertia of MGs. In real practice, sufficient system inertia is crucial to ensure a reliable power supply and stabilize the operation of MGs. Up to these days, MG inertia is decreasing due to the CIGs utilization. Even though several kinds of RES-based CIGs do have rotating elements like WTGs, their dynamics are decoupled by static power converters. Thus, the inertia property from CIG units is almost zero, resulting in the low inertia property. Any operational changes can cause lower damping, higher frequency oscillation, and faster rate of change of frequency (RoCoF), which are new challenges to maintain acceptable performance, stability, and resiliency of MGs. Specifically, MG inertia has been low during periods of low demand and high levels of RES-based CIGs. In the future, MGs will contain a substantial number of CIGs, which can lead to less stable MG performance and security than ever.

Natural oscillations with low frequencies are a typical characteristic in today's power systems, which can be triggered by occasional disturbances, e.g., sudden load changes and generation/line trips. If the system has adequate damping, such disturbances can be attenuated quickly. Especially in today's MGs, wind shear and tower shadow effects, interactions among phase lock-loop, high-level controllers and inner control loops from CIGs, malfunctions in generator exciters, and sensor feedback limitations may provoke forced oscillations (FOs). Unlike natural oscillations, FOs are triggered by such periodic external components. Despite sufficient damping in MGs, the continuous periodical perturbation introduces substantial FOs, which remain indefinitely until the sources are removed by a protection device or system operator. When the FOs interact with the existing natural oscillation modes, they can proliferate across the whole system with extremely magnified energy, resulting in power fluctuations, equipment damage,

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and limitation of power transfer capability [1]. In the Russian power grid, nine of ten turbines from the Sayano-Shushenskaya power station malfunctioned due to the FO event, leading to the power plant shutdown for three years [2]. In the North American power grid, the significant FO of 0.25 Hz occurred in the Eastern Interconnection (EI) due to power imbalance from a turbine unit, and this FO could be observed across the whole EI system with the natural oscillation mode of 0.24–0.25 Hz [3]. Technically, the intercept values of the steam turbine closed and reopened periodically for about four seconds, leading to widespread power oscillations (approximately 0.25 Hz) and generation trips.

Nowadays, the FO issue has emerged in various power systems around the world [4], and it is highly required by system operators to mitigate such effects as soon as possible. Thus, several approaches have been introduced to detect FO sources [5]. The dissolving energy flow approach is developed in [6,7] to determine the generator parameter uncertainties. In [6], the Bayesian technique is applied to deal with the generator parameter uncertainties. To enhance system operator's visibility throughout the vast grid network, the phasor data concentrators (PDCs) in the form of phasor measurement units (PMUs) are applied to locate the FO sources. Energy injection from generators and energy flow direction from PMUs are utilized for evaluating FO sources at the bus and equipment levels [7]. To reduce the information transfer, the PMUs are utilized in a distributed manner for determining the FO sources [8]. However, the limited coverage of PMUs and inadequacy of coordination among system operators have posed challenges in dealing with the FO source. It usually requires some time to discover the precise FO source, particularly when the FO occurs outside the control area of a system operator.

To solve this challenge, an effective control strategy through utility-scale can be mobilized to suppress the FOs before the source is identified and eliminated. This approach will provide appropriate time for system operators to locate the FO source without threatening system stability and resiliency. Up to now, a few approaches have developed the FO damping control methodologies. The authors have designed the damping control approaches based on generators [4,9], static synchronous compensators with energy storage systems (ESSs) [10], and modulation of power electronic converters [11] to reduce the FO effects. Nowadays, ESSs have become popular in various areas of utility grids due to their fast active power modulation capability, and they are preferred to be used for supporting the operations of WTG and PV-based CIGs. Together with the increasing RES-based CIG penetration (especially in MGs), ESSs are significantly utilized for the effective role of virtual inertia control, known as a virtual synchronous generator (VSG). During the low-inertia MGs (i.e., high portions of RES-based CIGs), the VSGs can emulate sufficient inertia power with damping to improve the grid stability with respect to frequency and voltage regulation [12–18]. In the existing literature, VSGs have been proposed to overcome low-frequency natural oscillations, but not the FOs. Moreover, none of the literature has been well established for investigating the FO effects in low-inertia MGs with high portions of RES-based CIGs.

Thus, this research introduces the extended VSG control application to arrest the FO effects corresponding to low-inertia MG conditions effectively. Comprehensive FO characteristics have been examined in time and frequency domains under the presence of a low-inertia MG with high levels of RES-based CIGs. The obtained results reveal the advantages of the extended VSG control in a low-inertia MG. The main contribution of this paper can be summarized as follows:

i) Comprehensive analysis of pre-, during-, and post-FO impacts has been demonstrated. The results clearly show how the FOs change the characteristics of the dominant modes under various MG operating points and FO conditions.
ii) To overcome the FO impacts, the extended VSG method is developed with respect to the emulation of virtual forced components. It can substantially diminish the severity of FO behaviors, where the FO magnitude/energy is effectively arrested within a safe level. This advantage could provide system operators adequate time to identify the exact FO source, leading to the correct elimination from MGs.
iii) Since the proposed approach relies on the VSG topology, none of the power measurements, e.g., PCDs and PMUs are required, reducing the complexity of implementation, and filed application. Compared with the normal ESS droop control and classical VSG control, the extended VSG technique provides outstanding benefits in a low-inertia MG for the sake of inertia control and further FO suppression.

The remainder of this paper is arranged as follows: Section 2 introduces a brief overview of frequency regulation-based inertia control. Section 3 models the MG under typical FO effects. Then, the extended VSG control to mitigate the FOs is proposed. Section 4 presents the case study. Finally, the concluding remark is provided in Section 5.

2. Definition of inertia and rate of change

MGs commonly contain a few synchronous generators (SGs), loads, and several CIG components (e.g., WTGs, PVs, and ESSs). Thus, the inertia constant of MG (H_{MG}) can be defined as [19–21]:

$$H_{MG} = \frac{1}{S_{MG}}\left[\sum_{i=1}^{n} H_{SG,i} \bullet S_{SG,i} + \sum_{i=1}^{m} H_{CIG,i} \bullet S_{CIG,i}\right] \tag{1}$$

where $H_{SG,i}$ is the inertia constant of the ith SG, $S_{SG,i}$ is the nominal power of the ith SG, $H_{CIG,i}$ is the inertia constant of the ith CIG, $S_{CIG,i}$ is the nominal power of the ith CIG, S_{MG} is the nominal apparent power of the MG, n is the number of SGs in the MG, and m is the number of CIGs in the MG.

The output frequencies of all CIG components can asymptotically reach the global frequency reference, where precise active power sharing can be addressed. Thus, the frequency inertia of CIGs can be improved by constraining the scope of the RoCoF of each CIG. Consequently, the initial variables of RoCoF from CIGs in this study are supposed to be finite constraints as [$RoCoF_{min}$, $RoCoF_{max}$].

In addition to initial variables, if the RoCoF of each CIG can be considered as [21]:

$$RoCoF_{min} \le RoCoF_i(t) \le RoCoF_{max} \forall i \in N \tag{2}$$

where $RoCoF_i$ is the changing rate of frequency of the ith CIG, $RoCoF_{min}$ is the minimum changing of frequency, N is the number of CIGs, and $RoCoF_{max}$ is the maximum changing of frequency.

Then, the inertia of each CIG and the entire MG can reach minimum values.

For common generators (e.g., SGs) in MGs, the frequency control can be defined based on the well-known swing equation as [19–22]:

$$2S_{SG}\frac{H}{f_n} \bullet \frac{df}{dt} = P_m - P_e - D(f_{SG} - f_n) \tag{3}$$

where S_{SG} is the nominal SG power, P_m is the mechanical power, P_e is the electrical (load) power, f_{SG} is the SG frequency, f_n is the reference frequency, H is the inertia constant, df/dt is the RoCoF, and D is the damping coefficient.

Considering (3), when H is variable, the inertia constant is inversely corresponding to the RoCoF. Subsequently, with (2), the minimum inertia constant of CIG (H_{CIG,i_min}) can be defined as [19–21]:

$$H_{CIG,i_min} = \frac{f_n}{2 \bullet RoCoF_{max} \bullet S_{CIG,i}}\left(\Delta P_{CIG,i} + D\Delta f_{CIG,i}\right) \tag{4}$$

where $\Delta P_{CIG,i}$ is the maximum (positive) active power deviation of ith CIG, $\Delta f_{CIG,i}$ is the maximum (positive) frequency deviation of ith CIG.

Later, the inertia constant of the ith CIG fulfills [21] as:

$$H_{CIG,i} \ge H_{CIG,i_min} \forall i \in N \tag{5}$$

Finally, the inertia constant of the entire MG can be considered as [19–21]:

$$H_{MG} \geq \frac{f_n}{2S_{MG}RoCoF_{max}} \sum_{i=1}^{n} \left(\Delta P_{CIG,i} + D\Delta f_{CIG,i}\right) \tag{6}$$

Therefore, the (6) is correct and can be examined by a proportional between frequency and power deviations corresponding to the $RoCoF_{max}$.

3. Forced oscillations in microgrids

3.1. Modeling of microgrid with forced oscillation

As shown in Fig. 1, the MG usually has a highly non-linear and time-varying. Thus, in the time-domain (t), the MG dynamics, including CIGs and FOs consist of differential–algebraic equations ($\dot{x}$), which can be written under the non-linear equations as:

$$\dot{x}(t) = g_x(x(t), u(t), d(\varepsilon t_d)) \tag{7}$$

$$\begin{bmatrix} \Delta\dot{x}(t) \\ \Delta\dot{u}^c(t) \end{bmatrix} = (A - B^c K C)\begin{bmatrix} \Delta x(t) \\ \Delta u^c(t) \end{bmatrix} + B^e \Delta w(t) + B^d \Delta d(t_d)$$
$$\approx (A^{'} - B^{c^{'}} K C)\begin{bmatrix} \Delta x(t) \\ \Delta u^c(t) \end{bmatrix} + B^e \Delta w(t), \tag{12}$$

$$y(t) = g_y(x(t), u(t), d(\varepsilon t_d)) \tag{8}$$

where t is the time, x and y are the state and output variables, respectively, u represents the input vectors from controllers, d is the forced disturbances, t_d is the vector of disturbance duration, ε is the even-triggered vector of d, and $g_x(.)$ and $g_y(.)$ represent the non-linear functions of x and y, respectively.

Considering the small-signal stability analysis, the study of FOs can be linearized at any op^{th} MG operating point. Let $(x - x_{op}) = \Delta x$ and $(d - d_{op}) = \Delta d$, where Δ is the small deviation. Accordingly, we have,

$$\Delta\dot{x}(t) = A\Delta x(t) + B^c \Delta u^c(t) + B^d \Delta d(t_d) + B^e \Delta w(t) \tag{9}$$

$$\Delta y(t) = C\Delta x(t) \tag{10}$$

where $A \in R^{nxn}$ and $C \in R^{n_c xn}$ are the state and output matrices, $B^c \in R^{nxn_c}$, $B^d \in R^{nxn_d}$, and $B^e \in R^{nxn_e}$ are respectively the input and feed-forward matrices associated with MG control signals and forced disturbances (FDs), w is the input vector of external disturbances from RES and load units, and u^c and d are the input vector from the VSG controllers and FDs, respectively.

To study the FO behaviors, the CIGs are equipped with the VSG control, and the FDs, i.e., $d(t_d)$ are usually represented by a combination of sine wave components as:

$$d(t_d) = \sum_{i=1}^{N_i} \left(P_{d,i}\cos\left(2\pi f_{d,i} t_{d,i} + \theta_{d,i}\right)\right) \tag{11}$$

where $i = 1, 2, \ldots, N_i$ is the counter of sine components or FDs, N_i is the total number of sine components in the MG, P_d, f_d, and θ_d are the vectors of disturbances amplitudes, frequencies, and phases, respectively.

At any op^{th} operating point in which Δ is very small, it is found that the FDs can be assumed to be constant, therefore $B^d \Delta d(t_d) \approx B^d d(t_d)/dt \approx \Psi \in R^{nxn_d}$, where Ψ is the constant matrix. Considering the following assumptions: *i*) if the FDs excite any state, output, or other algebraic variables (i.e., $\{x_{alg}, y\} \in x$) and *ii*) if the FDs do not create additional state variables to the MG, thus the corresponding matrices are affected by the FDs such that $B^d \Delta d(t_d) \in \{A(:,k), B^c(:,k)\}$, where k is the column in which the FDs are injected. As a result, the closed loop of (9) and (10) can be rewritten by the following equations,

when : $A \xrightarrow{A(:,k)+\Psi} A^{'}$, $B^c \xrightarrow{B^c(:,k)+\Psi} B^{c^{'}}$, (13)

where $A^{'}$, $B^{'}c$, and $C^{'}$ are respectively the excited A, B, and C by Δd, and K is the CIG-based VSG controller.

In (12), the eigenvalues (λ_j) of the closed-loop system $A_{cl} = (A^{'} - B^{c^{'}} KC)$ can be calculated by: $det(A_{cl} - \lambda_j I) = 0$ and $\lambda_j = -\sigma_j \pm j2\pi f_j$, where $j = 1, \ldots, N_j$ is the counter index of system modes (i.e., λ_j, and σ_j), N_j is the total number of system modes, σ_j, and f_j are the real part and frequency of λ_j, respectively. Accordingly, the damping ratios of $\lambda_j(\zeta_j)$ are obtained by:

$$\zeta_j = -\frac{\Re(\lambda_j)}{\sqrt{(\Re(\lambda_j))^2 + (\Im(\lambda_j))^2}} = -\frac{\sigma_j}{\sqrt{(\sigma_j)^2 + (2\pi f_j)^2}} \tag{14}$$

The MG is very sensitive to operating changes and FDs due to the low inertia and intermittent output of RES-based CIGs. Thus, the participation factors of λ_j are uncertain and they have a chance to participate with the state variables of CIGs. As a result, we can simply find the dominant oscillation modes (λ_{do}) by:

$$\lambda_j = \lambda_j(\min(\zeta_j)) \tag{15}$$

Thus, the dominant frequency (f_{do}) can be found by:

$$f_{do} = f_j(\min(\zeta_j)) \tag{16}$$

where min represents the minimum value.

In the low-inertia MGs with CIGs, if one or more of these conditions *i*) eigenvalues of the closed-loop system A_{cl} are critical, *ii*) P_d is high enough, or *iii*) f_d is close or equal to f_{do}, is/are met, the FOs are then observable in the measured output y. Accordingly, they will exhibit the FO responses interacting with multiple modes in the following form as:

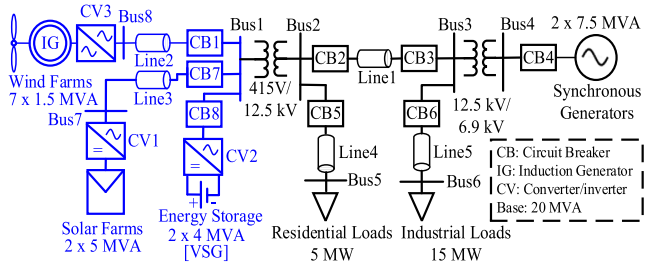


Fig. 1. The studied stand-alone MG with RES-based CIGs.

$$y(t) = \sum_{j=1}^{N_j} \left(\Lambda_j A_j (\cos(2\pi f_j t)) e^{-\zeta_j t}\right) + \sum_{i=1}^{N_i} \left(\Lambda_{d,i} P_{d,i} (\cos(2\pi f_{d,i} t_{d,i} + \theta_{d,i}))\right) \quad (17)$$

where Λ and Λ_d are the modal amplification factor of the first and second terms of (17).

3.2. Extended VSG control for FO suppression

In addition to the VSG concept, if the active power through the power electronics converters (e.g., CIGs) of ESSs is controlled proportionally with the derivative of the MG frequency, virtual inertia in the system can be properly emulated, which will contribute to improving the inertial response of the MG against disturbances. The basic control law for the active power of the power converter can be expressed as [12–18]:

$$P_{emulate} = k_a \bullet f_n \frac{d(\Delta f)}{dt} \quad (18)$$

where $P_{emulate}$ is the emulated (active) power, and k_a is the proportional conversion gain.

Owing to very low oscillation amplitudes in MGs, the response of the modes with high values of ζ_j may not be observed in (17). In addition, the well-damped modes will not create further stability issues. Therefore, the poorly damped modes, which are patent in the signal (17), are sufficient to address the VSG control. As proofed in [12–18], the VSG control can be applied to regulate the frequency fluctuation during high levels of RES-based CIG penetration. In the study, we extend the VSG concept to include the FO impacts from RES-based CIG components. As illustrated in Fig. 2, the extended VSG control first calculates the RoCoF based on the derivative technique following the disturbances, including FDs. Then, the additional inertia power with sufficient damping can be emulated by injecting the VSG control signal into the RES-based CIG active power control loop. This study also assumed that the ESS provides sufficient energy for emulating inertia against disturbances and FOs. In addition to the derivation control, it is sensitive to noise. Thus, a low-pass filter could be added to eliminate the noise from the frequency measurement.

Let $\Delta\bar{f}$ be the measurement of the mean frequency deviation, which contains high observability of disturbances (i.e., RES/load changes and FDs). We can write the output signal $y = \Delta\bar{f}$ in the form of (17). The $\Delta u^c(s) = \Delta\bar{f}(s)(J_{VSG} \bullet s + D_{VSG}) + \widetilde{d}(s)$ is the stabilizing power change from the VSG controller, which is feedbacked to the MG or RES-based CIG systems. As a result, the dynamic transfer function of the proposed VSG control can be derived in the frequency domain (s) as [20,23,24]:

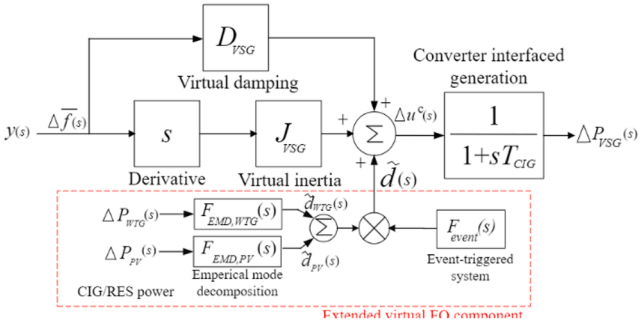


Fig. 2. The structure of VSG control with extended virtual FO component.

$$\Delta P_{VSG}(s) = \frac{\Delta u^c(s)}{1 + sT_{CIG}} \quad (19)$$

where J_{VSG} and D_{VSG} are the virtual inertia and damping constants, respectively, and T_{CIG} is the time constant of the CIG converter.

The estimated value of $\widetilde{d}(s)$ can be determined from the measured signal of the penetrated CIG power via the empirical mode decomposition (EMD) technique [25–27]. Considering the extension of virtual component-based FDs (see Fig. 2), the EMD function, i.e., $F_{EMD,WTG}(s)$, $F_{EMD,PV}(s)$ could effectively extract the FD component from the CIG power signals, i.e., $\Delta P_{WTG}(s)$ and $\Delta P_{PV}(s)$, which is then used to emulate the virtual forced component based on an even-triggered system, i.e., $F_{event}(s) \in \{0 \quad 1\}$ to the VSG unit, resulting in the reduction of dynamic FD effects. Thus, we have $\widetilde{d}(s) = F_{event}(s) \bullet [\Delta P_{WTG}(s) \bullet F_{EMD,WTG}(s) + \Delta P_{PV}(s) \cdot F_{EMD,PV}(s)]$.

In addition, during the FO event, the even-triggered system enables the extended function of $\widetilde{d}(s)$ passing through the VSG to directly eliminate the FD components. When there is no FO event, the $\widetilde{d}(s)$ is deactivated by the even-triggered system, where the normal function of VSG control remains active to attenuate the low inertia and damping effects in the existing MG.

As clearly seen in (12), it is possible to compute the ζ_j under FD conditions. From (19), the inertia and damping of the MG containing sufficient forced components can be imitated to suppress the frequency fluctuation with existing FO effects by appropriately adjusting the values of J_{VSG} and D_{VSG} corresponding to the virtual forced component based on an even-triggered system to increase the ζ_j.

4. Simulation results and discussions

The simulation results and discussion are characterized in seven subsections as follows: *A*) analysis of pre-, during-, and post-FO impacts on a dominant mode of low-inertia MG; *B*) effectiveness of virtual forced component in an extended VSG; *C*) analysis of VSG parameters against FO suppression; *D*) VSG efficiency under low-inertia MG with multiple FDs; *E*) VSG robustness in the presence of FD-based generation trip; *F*) VSG robustness in the four-area interconnected MG; and *G*) VSG sensitivity against RoCoF, inertia reduction, and RES penetration with FD impact. The MG test models with simulation waveforms are built in the MATLAB/Simulink® 2021 environment via a personal computer with installed RAM 16.0 GB, 11th Gen. Intel® Core i5-11400F@2.60 GHz.

4.1. FO effect on dominant mode of low-inertia MG

In this subsection, we shall first demonstrate how the FO changes the system modes' characteristics through time-varying. Considering (17), the signal y exhibits multiple components in which the first and second terms are the components of the MG modes and FOs, respectively. If the disturbance frequency (f_d) is close to any system modes, the $d(t_d)$ may lead to the corresponding twin components, i.e., system mode (first term) and forced (second term) components. The terms Λ, A, and Λ_d, P_d significantly depend on system operating points, system conditions, and measured locations of $y(t)$. Although the signal decomposition of (17) is useful for FO detection and further identification, the impacts of FO do not include the frequencies and damping ratios of system modes. Thus, (17) can be analyzed by three major oscillation periods:

$$y(t) = \begin{cases} y_{pre}(t), \text{if } t = [t_0 \quad t_d) \\ y_{du}(t), \quad \text{if } t = t_d \\ y_{pos}(t), \text{if } t = (t_d \quad t_d'] \end{cases} \quad (20)$$

$$y_{pre}(t) = \sum_{j=1}^{N_j} \left(\Lambda_j A_j (\cos(2\pi f_j t)) e^{-\zeta_j t}\right) \quad (21)$$

$$y_{du}(t) = \sum_{j=1}^{N_j} \left(\widetilde{A}_j (\cos(2\pi \widetilde{f}_j t) \right) e^{-\widetilde{\zeta}_j t} \tag{22}$$

$$y_{pos}(t) = \sum_{j=1}^{N_j} \left(A_j^{'} (\cos(2\pi f_j^{'} t) \right) e^{-\zeta_j^{'} t} \tag{23}$$

when

$$\widetilde{f}_j = \widetilde{\epsilon}_j(d) \bullet f_j, \widetilde{\zeta}_j = \widetilde{\beta}_j(d) \bullet \zeta_j \tag{24}$$

$$f_j^{'} = \epsilon_j^{'}(d) \bullet f_j, \zeta_j^{'} = \beta_j^{'}(d) \bullet \zeta_j \tag{25}$$

where the subscriptions of *pre*, *du*, and *pos* mean the pre-, during-, and post FO-events, respectively. The terms of $\left(\widetilde{A}\widetilde{f}, \widetilde{\zeta}\right)$ and $(A^{'}, f^{'}, \zeta^{'})$ are respectively the corresponding variables during- and post-FO events, $\widetilde{\epsilon}(.)$ and $\beta^{'}(.)$ are the affected factors by $d(t_d)$ on the frequencies and damping ratios of system modes, respectively.

In this subsection, we define the effects of during and post-FO events on the oscillation frequencies and damping ratios of system modes at any j^{th} system mode and i^{th} forced disturbance, i.e., $\widetilde{E}_{fr}$, $\widetilde{E}_{da}$, $E_{fr}^{'}$, and $E_{da}^{'}$ as,

$$\widetilde{E}_{fr} = \left| \widetilde{f}_{s,j} - f_{s,j} \right| \tag{26}$$

$$\widetilde{E}_{da} = \widetilde{\zeta}_{s,j} - \zeta_{s,j} \tag{27}$$

$$E_{fr}^{'} = \left| f_{s,j}^{'} - f_{s,j} \right| \tag{28}$$

$$E_{da}^{'} = \zeta_{s,j}^{'} - \zeta_{s,j} \tag{29}$$

The indices $\widetilde{E}_{fr}$ and $E_{fr}^{'}$ respectively mean the absolute differences between dominant system frequencies during- and post-FO events, while the indices $\widetilde{E}_{da}$ and $E_{da}^{'}$ signify how the damping ratios of the dominant modes are altered during- and post-FO events.

At the normal operating point, the dominant mode in this system is a mode with damping and frequency around 40.1 % and 0.361 Hz, respectively. The mode is mainly associated with two SGs. At the normal operating point ($H_{\mathrm{MG}} = 100$ %), the participation factor of this mode is accounted for 93.37 %, associated with the rotor speeds and phase angles of SGs, while the rest percentage is associated with the state variables of RES-based CIGs. When the inertia decreases to zero with fine steps to the high penetrations of RES-based CIGs, the frequencies of the dominant mode are respectively altered from 0.361 to 0.909 Hz, while the damping is either increased or decreased in the range of 40.1 % to 1.85 %. The participation factor of this mode is prone to be mainly associated with the RES-based CIGs state variables under the desecration of H_{MG}.

The malfunction can trigger the FO among converter control loops in the converter of WTG or PV systems. In this scenario, the FO is assumed to be 1 % of the PV output (P_{PV}) as: $P_{PV} = P_{PV}(t) + 0.01\ P_{PV}\ (t_d)\cos(2\pi f_d t_d)$. The duration for pre-, during-, and post-FO events are set at [0 20), [20 80], and (80 100] s, respectively. In the pre-FO event, the MG response is fluctuated by ambient and small RES/load changes. During the FO event, the MG is oscillated by ambient, small RES/load change, and $d(t_d)$. After the disappearance of FO, the MG is perturbated by ambient, small RES/load changes and the consequent effect of $d(t_d)$. Fig. 3 shows the transient responses of the MG frequency deviation. By conducting the small-signal stability analysis, the f_s and ζ_s are obtained. Accordingly, the damping ratios and frequencies of the dominant mode at $H_{MG} = 100$ %, 66.66 %, 33.33 %, and 0 % are: {0.401, 0.361 Hz}, {0.289, 0.476 Hz}, {0.131, 0.689 Hz}, and {0.018, 0.909 Hz}, respectively. In this case, the f_d is set at 0.05 Hz (lower than the dominant frequency) or 1.2 Hz (higher than the dominant frequency) as

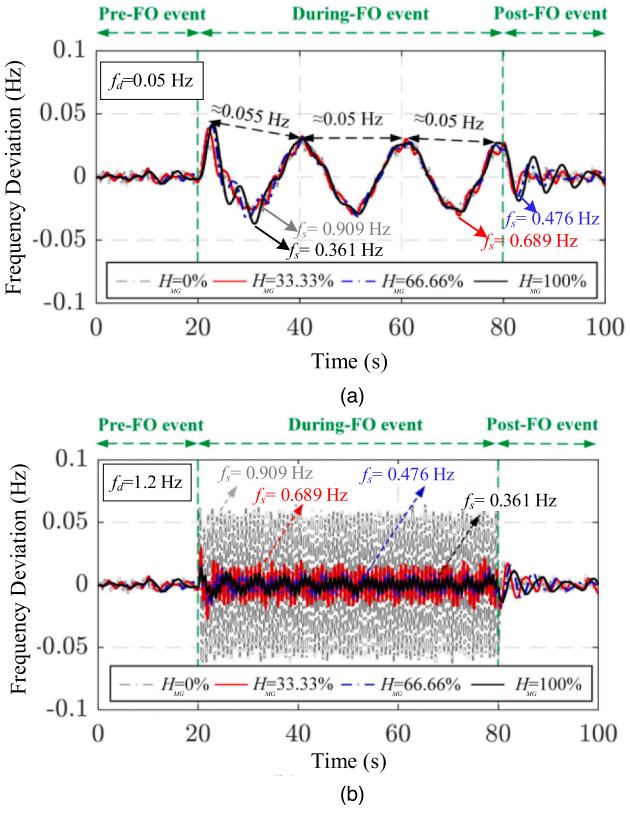


Fig. 3. MG responses under different f_d and H_{MG}; a) $f_d = 0.05$ Hz; b) $f_d = 1.2$ Hz.

demonstrated in Fig. 3(a) and Fig. 3(b), respectively. At $f_d = 0.05$ Hz, see in Fig. 3(a); in the during-FO event, the frequency deviation exhibits high oscillation amplitude with approximately 0.05 Hz oscillation frequency for all levels of H_{MG}. Thus, the system mode is completely dominated by the FO for all levels of H_{MG}. In the post-FO event, the frequency deviation shows the oscillation responses with the same frequency as the system mode. At $f_d = 1.2$ Hz as shown in Fig. 3(b); In the during-FO event, although the 1.2 Hz disturbance frequency does not resonate with the 0.909 Hz system mode when $H_{MG} = 0$, the frequency deviation clearly demonstrates the effect of FO on the system mode and the oscillation amplitude is extremely high compared to the others. As a result, the frequency deviation oscillates completely with 1.2 Hz frequency. As seen, the oscillation amplitudes are rapidly decreased when $H_{MG} = 33.33$, 66.66, and 100 %. When H_{MG} is increased, the frequency deviation exhibits both components of the FO and system mode. In this period, the FO effect is moderate, thus, the system mode tends to dominate the FO.

Fig. 4 illustrates the mesh maps of indices in (26)–(29) and maximum frequency deviation. When f_d is varied from 0.05 to 2 Hz, and H_{MG} is changed from 0 to 100 %. During and after the FO events, the FO includes in the frequency deviation, thus affecting the characteristics of the system mode. The indices in (26)–(29) are estimated by the subspace state-space identification known as 4*SID* [28]. In these states, the FO effect includes in the measured signals, where $\widetilde{f}_s$, $\widetilde{\zeta}_s$, $f_s^{'}$, and $\zeta_s^{'}$ are estimated by the 4*SID*. In Fig. 4(a) and 4(d), the FO affects the dominant mode frequency at all disturbance frequencies, especially during the FO event with very low H_{MG} (less than 30 %). When f_d is a small value below 0.9 Hz, the FO dominates the system mode. However, the FO does not affect the dominant mode frequency after the FO event. As observed in Fig. 4(b), the damping ratio of the system mode is significantly and periodically altered from the steady state during the FO event at all levels of H_{MG}. After the FO event, see Fig. 4(e), the FO effect on damping is periodic and miniature. As seen in Fig. 4(c), the maximum frequency

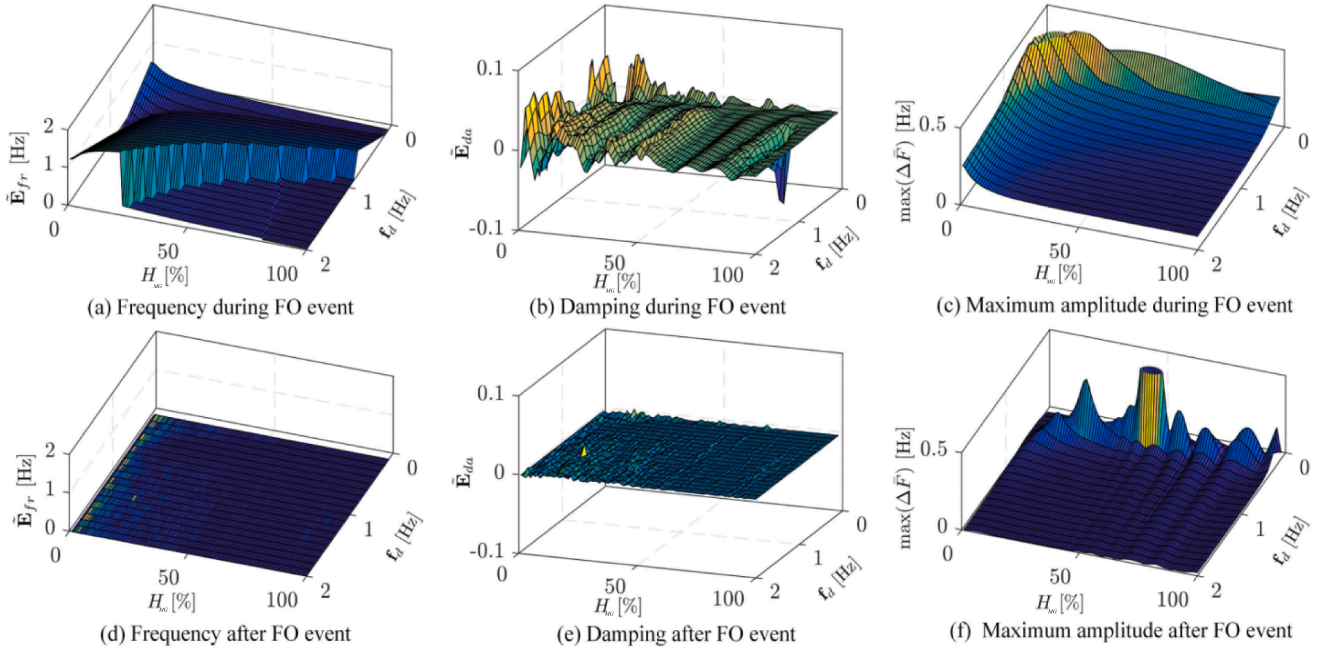


Fig. 4. Mesh maps of $\widetilde{E}_{fr}$, $\widetilde{E}_{da}$, E'_{fr}, and E'_{da} and maximum oscillation amplitude under f_d and H_{MG}.

deviation is high when f_d and H_{MG} are low. In this case, during the FO, the problem of frequency deviation is intensified by both low inertia and resonant effect between f_d and f_s, which is in the range of 0.361 to 0.909 Hz. Although the oscillation amplitude is decreased when f_d and H_{MG} are increased, the frequency deviation may exceed the acceptable range of the frequency operating standard (±0.2 Hz) [29]. As shown in Fig. 4(f), the frequency deviation is high when f_d is low. The reasons can be *i*) the same as that of the case of the during-FO event and *ii*) the consequent effect of the FO.

Thus, in a variable low-inertia MG, the FO effect over the dominant mode can be signified as;

i) When the FO occurs (identified in the time domain), one can regard a forced disturbance as an internal input. Accordingly, the FO effect on system mode can be investigated. The FO effect is periodic, while the severity and behavior of the FO significantly depend on the system inertia and disturbance frequency.
ii) The FO can exhibit an identical oscillation response to the FD characteristics if the system inertia is close to zero. During the FO event, the FO response is not always influenced by the system dominant mode. Irrespective of the closeness of the dominant frequency, the FO effect on damping is miniature, but there is no significant impact on the mode frequency. Hence, the FO can affect the damping and frequency of the system mode during and after the FO events significantly.

To effectively arrest the FO effects corresponding to low-inertia MG conditions, the extended VSG control application is investigated in the following subsection.

4.2. Effectiveness of virtual forced component in extended VSG

This section demonstrates the contribution of the virtual forced component in the extended VSG control. At the operating point, the total inertia of the MG is reduced to 50 % of its nominal value ($H_{MG} = 50$ %), corresponding to the disturbance events of sudden load and RES change with power amplitudes of −0.2p.u. and + 0.2p.u., respectively. The dominant mode of MG has the frequency (f_s) around 0.59 Hz with a damping of 17 %, which corresponds to the state variable of SGs. The MG contains the periodic FD with an amplitude of 2 % as: $d(t_d) = 0.02 P_{wind}(t_d)\cos(2\pi f_d t_d)$. While the f_d resonates at the same frequency as the existing f_s ($f_d \approx 0.59$ Hz).

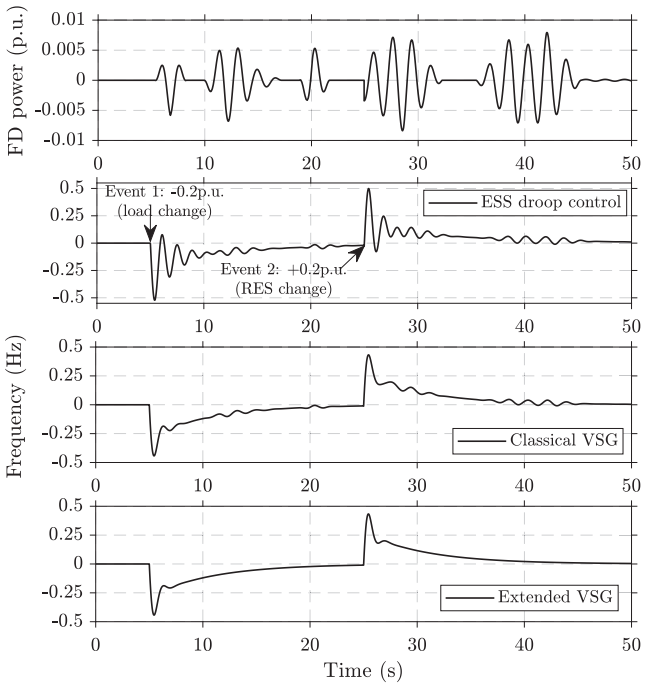


Fig. 5. Frequency response of the MG in the extension of virtual forced emulation.

Following the applied disturbance events (see Fig. 5), it is clearly seen that the FD power exacerbates in the MG frequency with the ESS droop control. It significantly causes further frequency oscillations with high amplitudes during the stabilizing time of the MG. By applying the classical VSG control, the effect of frequency oscillations triggered by

the FD component is deteriorated, while small frequency oscillations could be observed. In addition, the FD effect is weakened by the additional gains of virtual inertia and damping of classical VSG control. By applying the extended VSG control, the small frequency oscillations caused by the FD are successfully diminished due to the extension of the virtual forced component. In fact, the emulation of the virtual forced component effectively deducts the actual FD component, leading to a smoother frequency deviation of the MG. Thus, it is confirmed that the extended VSG control could simultaneously diminish the existing FD effects and also augment the MG stability in a low-inertia condition.

4.3. Analysis of VSG control parameters over FO effect

To analyze the sensitivity of VSG control parameters against substantial FO impact, the trajectories of dominant eigenvalues are derived by varying the VSG control parameters (i.e., J_{VSG} and D_{VSG}). At the operating point, the total inertia of the MG is reduced to 70 % of its nominal value (H_{MG} = 70 %), corresponding to the small or medium penetration level of RES-based CIGs. In this condition, the dominant mode of MG has the f_s around 0.44 Hz with a damping of 29.6 %, which corresponds to the state variable of SGs. As seen in Fig. 6, the WTG-based CIG output contains the periodic FD type with a disturbance amplitude of 3 % of WTG output as: $P_{wind} = P_{wind}(t) + 0.03\ P_{wind}(t_d)\cos(2\pi f_d t_d)$. While the f_d resonates at the same frequency as the existing f_s ($f_d \approx 0.44$ Hz).

Fig. 7 displays the effect of the increasing J_{VSG} and D_{VSG} against FO suppression. The arrow signifies the increment of corresponding VSG control parameters. By increasing the J_{VSG} and D_{VSG}, it is obvious that the dominant conjugate eigenvalues move to the left side of the s-plane, resulting in a more stable and secure system.

The result of the eigenvalue analysis corresponds to the MG frequency response, as shown in Fig. 8. With small J_{VSG} and D_{VSG}, the MG frequency tremendously oscillates over ± 0.2 Hz (e.g., larger overshoots/drops), which endangers the acceptable range of the frequency operating standard for MGs [29]. Such an event can trigger system instability and activate protection devices, resulting in generation trips, load-shedding, and power blackouts. By increasing the J_{VSG} and D_{VSG}, the FD effect is effectively arrested, where the frequency excursion is smoother with lower frequency overshoot and nadir. Therefore, by employing the extended VSG control, the forced components are substantially arrested, resulting in superior MG operation and performance against the penetration of RES-based CIGs.

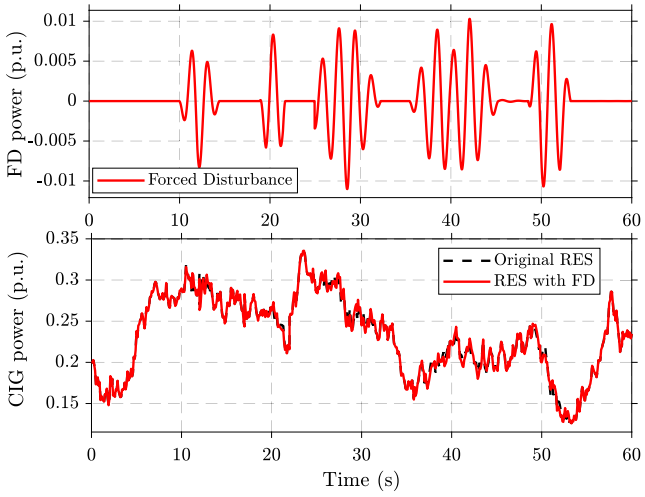


Fig. 6. WTG-based CIG power output containing 3 % of combined FD.

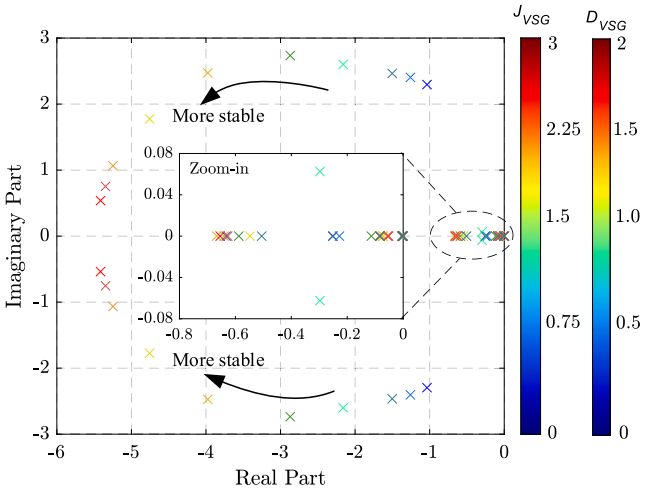


Fig. 7. Eigenvalue trajectory of the MG with FD (3 %) against the variation of VSG control parameters.

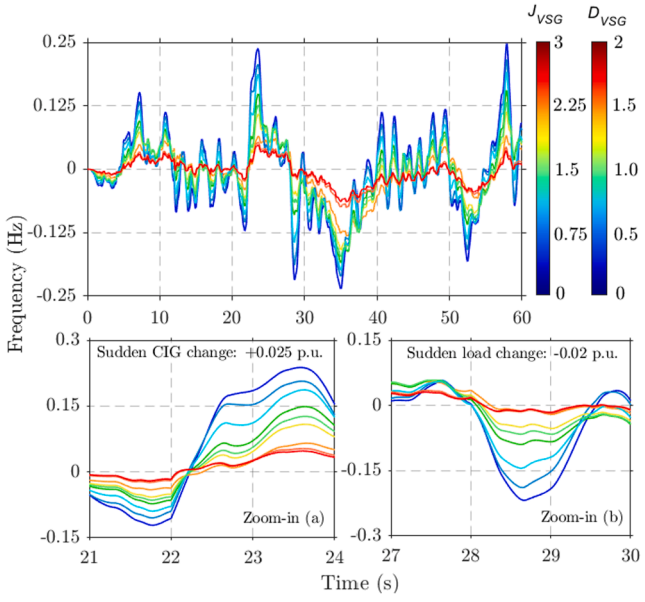


Fig. 8. MG frequency response with FD (3 %) against the variation of VSG control parameters.

4.4. VSG efficiency on low-inertia MG with multiple FO types

In this case, the VSG efficiency is investigated under the condition of low inertia with multiple FD types. In this condition, the dominant mode of the low-inertia MG has the f_s of around 0.58 Hz with a damping of 17.5 %, which corresponds to the state variable of SGs. The total system inertia is reduced to 50 % of its nominal value (H_{MG} = 50 %), signifying a low inertia condition. To force more severe operating conditions, the FDs are accounted for 5 % of the WTG power output as: $P_{wind} = P_{wind}(t) + 0.05\ P_{wind}(t_d)\cos(2\pi f_d t_d)$. To clearly demonstrate the original effect of the low-inertia MG, the FD does not exist at the beginning period between t = [0 20] s. Then, at the periods t = [20 60] s, t = [60 100] s, t = [100 120] s, and t = [120 140] s, *i*) periodic FD with f_d of 0.58 Hz, *ii*) combined FD with f_d of 0.425, 0.58, 0.725 Hz, *iii*) pure sine FD with f_d of 0.58 Hz, and *iv*) high-frequency FD with f_d of 1.25 Hz contain in the WTG power output, respectively. The J_{VSG} and D_{VSG} are set at 1.65 and 1.20, respectively. The proposed method is also compared with the well-known power frequency (*P-f*) droop controller based on ESS operation [33–35].

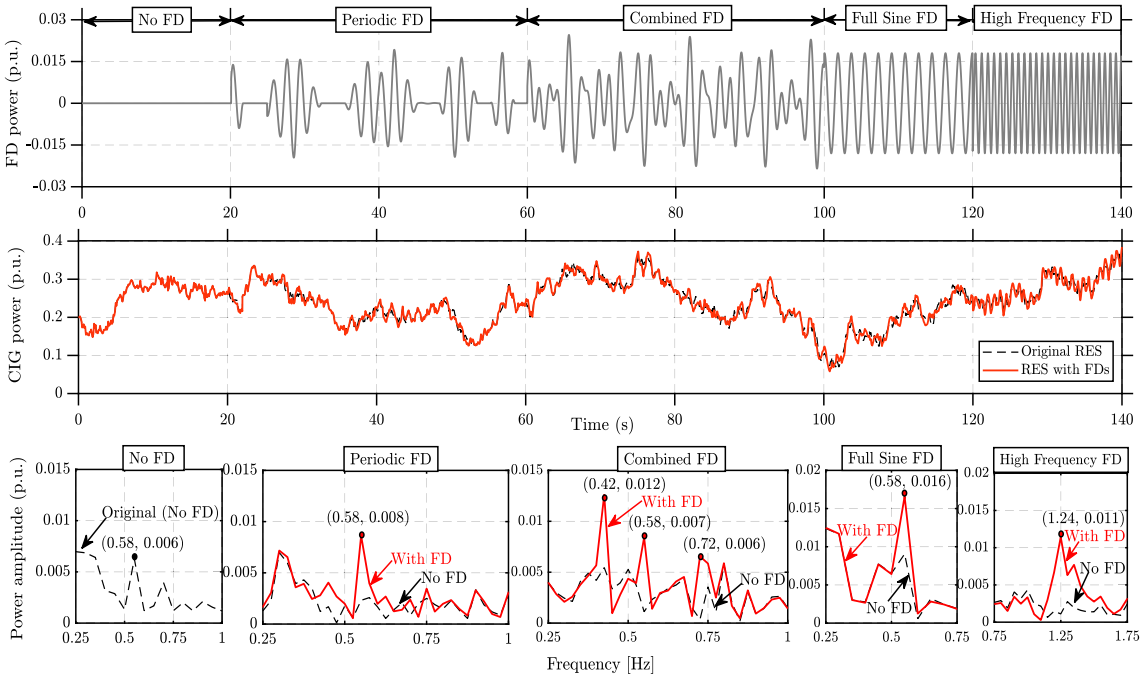


Fig. 9. The output of WTG-based CIG containing 5 % of the multiple FDs.

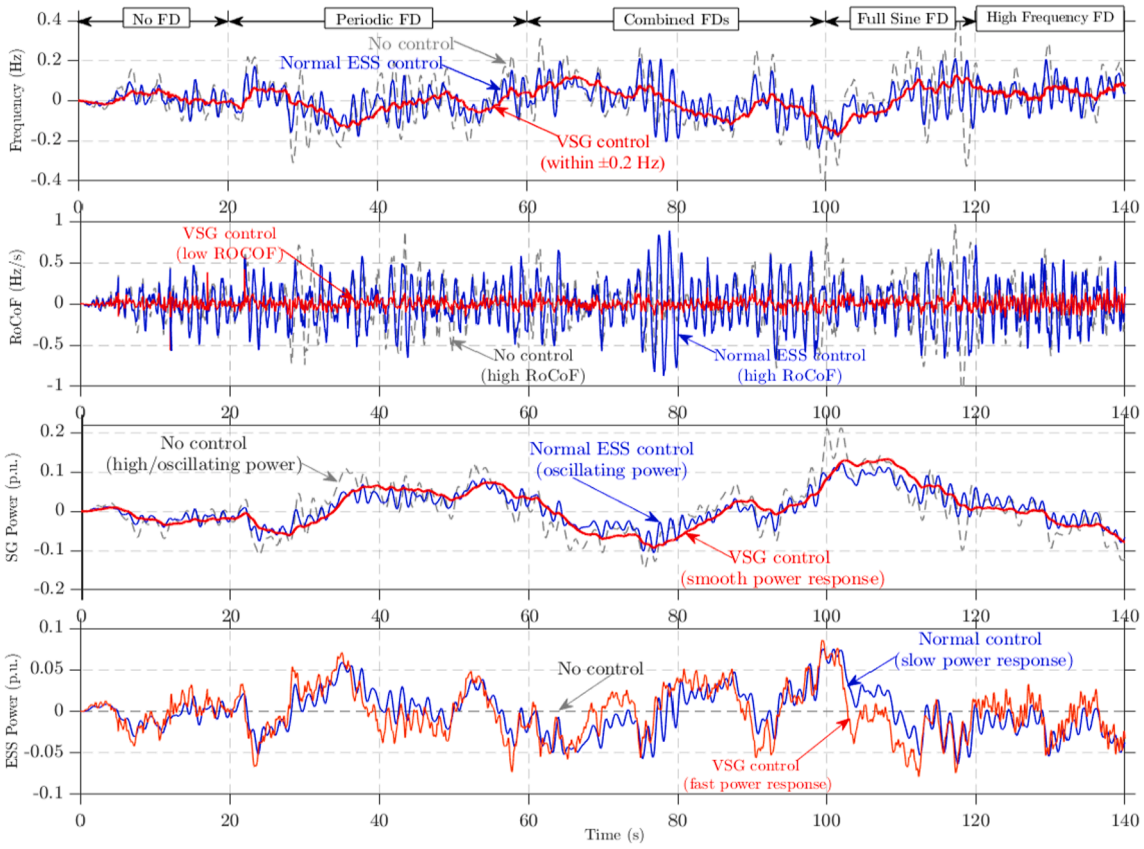


Fig. 10. MG response under low inertia with multiple FDs.

As observed in Fig. 9, the FO source is still obscured by the fluctuation of WTG-based CIG output when the FD size is increased. By conducting the FFT analysis, it divulges significant FD amplitudes in the frequency domain. This analysis signifies that all poor FD sources can contribute to the high oscillation amplitudes.

Fig. 10 illustrates the MG response under low inertia with multiple FD conditions. When the FD does not exist in the WTG output, the frequency deviation of the MG can be maintained within the normal frequency operating standard of ± 0.2 Hz by all approaches (i.e., no controller, normal ESS droop control, and VSG control), corresponding

to the lower RoCoF. After $t > 20$ s, when the FDs exacerbate in the MG, high-frequency deviations (over ± 0.2 Hz) with high RoCoF in the cases of no control and normal ESS control can be observed. These control approaches result in MG instability due to insufficient inertia and damping. Conversely, the additional inertia with damping is imitated by the proposed VSG control, leading to the effective FO suppression. It is evidently seen that the VSG control exhibits a faster power response than normal ESS control, resulting in a lower RoCoF and frequency deviation. Consequently, a smoother power deviation from the SG could be observed, leading to less mechanical stress on the generator and turbine components. Thus, the stability margin of the low-inertia MG could be significantly augmented.

4.5. VSG robustness under FO-dominated generation trip

During the FDs, a generator could be tripped by the protection devices or manually disconnected by a system operator, avoiding potential damages. Hence, the developed VSG method should be validated under such operating conditions of the FD-dominated generation trip.

In this case, the RES power output is penetrated at a high level (i.e., 70 % of their capacity), resulting in the reduction of H_{MG} to 30 % from its nominal value. Accordingly, the f_s increased to 0.72 Hz with the degraded damping of 8.35 %, corresponding to the state variable of SGs. At $t = 50$ s, the malfunction among converter control loops in the WTG converter occurs, emerging the forced characteristic. As displayed in Fig. 11, the FD has a disturbance amplitude of 5 % of the WTG power output as: $P_{wind} = P_{wind}(t) + 0.05\ P_{wind}(t_d)\cos(2\pi f_d t_d)$. While the f_d also resonates at the same frequency as the existing f_s ($f_d \approx 0.72$ Hz). Then, the wind farm is detected as the FO source by an FO identification technique. To completely eliminate the FO, the wind farm unit should be disconnected from the MG to prevent the MG instability and cascading failures.

Fig. 12 reveals the MG responses under the critical condition of the FD-dominated WTG trip. In the case of no control, the highest frequency (over ± 0.3 Hz) and RoCoF (over ± 1 Hz/s) oscillations could be vastly observed with longer setting time, while the MG completely loses its stability after the trip/disconnection of the FD source, leading to the cascading failure and blackout. This problem comes from the fact that the MG is operated in low inertia and damping conditions. In addition, the abrupt disconnection of the WTG triggers the larger transient in which the MG does not have the potential to sufficiently respond to such a system reconfiguration. Evidently, the FO source's disconnection significantly results in a higher RoCoF oscillation. In the case of normal ESS control, the disconnection still causes high frequency (over ± 0.2 Hz) and RoCoF (within ± 1 Hz) deviations with longer setting time. However, after the disconnection, the frequency deviation in the case of normal ESS control can be recovered to the acceptable range of frequency within ± 0.2 Hz with the RoCoF deviation of ± 0.5 Hz/s. This is because the ESS control could quickly respond to the mismatched power after the FO disconnection, avoiding MG instability and blackout. In the case of VSG control, the lowest frequency and RoCoF deviations could always be observed. Obviously, the FD source's disconnection is unnecessary since the VSG control could regulate the MG frequency effectively. On the one hand, the disconnection can also be applied in the case of VSG control, where the transient effect emerges with a smaller amplitude than those of no control and normal ESS control. In addition to the zoom-in view (a) of Fig. 12, the frequency drop could be significantly reduced from −0.6 Hz to −0.17 Hz, which is about 70 % reduction, where the RoCoF response is apparently increased; see the zoom-in view (b). Therefore, this case confirms the robustness of the VSG control to coordinate the control and network solutions for the FO elimination, while the trip/disconnection of the FO source could not significantly jeopardize the MG stability under the low inertia condition.

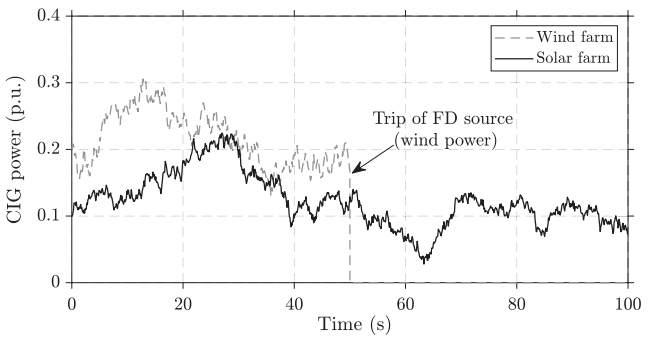


Fig. 11. RES-based CIG power output with a trip of the FD source.

4.6. VSG robustness under a four-area interconnected microgrid

In this section, the robustness of the extended VSG control is verified by extending to a multi-area MG based on four identical interconnected control areas with tie-lines (see Fig. 13). In this study, the area frequency deviation is individually used to control the VSG in its control area. The VSG power with virtual forced components can be exchanged to other areas via the tie-lines. At the initial condition, the WTG power with an FD amplitude of 3 % is penetrated to area 1 at a substantial level (i.e., 50 % of its capacity), reducing its inertia to 40 % from the nominal value. At 30 s, the PV power with an FD amplitude of 3 % is penetrated to area 4 at a substantial level (i.e., 25 % of its capacity), reducing its inertia to 70 % from the nominal value. The f_{d1} and f_{d4} resonate at 0.60 Hz, and 0.75 Hz, respectively. All power disturbances are depicted in Fig. 14.

As seen in Fig. 15, the system frequencies in all areas severely oscillate over the frequency operating standard (±0.2 Hz) in the case of no extended VSG control, i.e., the first scenario. A larger frequency transient (±0.5 Hz) could be observed in the areas of RES-based FDs, while a lower transient occurs in the areas of no FD sources, see Fig. 15, zoom in (a). Clearly, all system frequencies become unstable since the FDs have simultaneously been penetrated at 30 s due to the connection of the PV farm in area 4. In the second scenario, when the extended VSGs are applied to the areas of no FD sources (i.e., areas 2 and 4), the frequency transients and oscillations in such areas are significantly arrested within the acceptable ranges of ± 0.15 Hz, while the FD areas still possess the larger frequency transient and oscillations due to the existing FO components corresponding to the absence of VSG control, see Fig. 15, zoom in (b). However, it should be noted that the frequency oscillations in all areas of the second scenario are lower than in the first scenario due to the existence of VSG control in its area corresponding to other areas via transmission lines. In addition to the third scenario, when the extended VSGs are directly integrated into the FD areas, the emulation of the virtual forced component could effectively withhold the actual FO effects, leading to a huge reduction of frequency transients and oscillations in all interconnected areas. Obviously, frequencies in all areas oscillate less than ± 0.2 Hz during a transient event and less than ± 0.1 Hz during a steady-state event, which are within the acceptable range of the frequency operating standard; see Fig. 15, zoom in (c). Therefore, this section confirms that the extended VSG is necessary to maintain the stable and robust operation of the interconnected MG against the existence of the FO component corresponding to the low inertia MG environment.

In the complex MG structure, the frequency deviation of each node/area (e.g., Δf_1, Δf_2, Δf_3, Δf_4) in the MG may be different from the measurement of mean frequency deviation ($\Delta\bar{f}$) due to the complicated dynamic processes. Any signal processing and filtering introduce the delays that should be considered for emulating the VSG power to the MG. Typical filters on secondary frequency control and tie-line use about 2 s or more for the data acquisition and decision cycles [30]. In the control systems, it is well known that time delays can degrade the system's performance. Thus, in the extended VSG study, the communication delays (e^{-st_D}) may affect the VSG power generation via $\Delta\bar{f}$, where

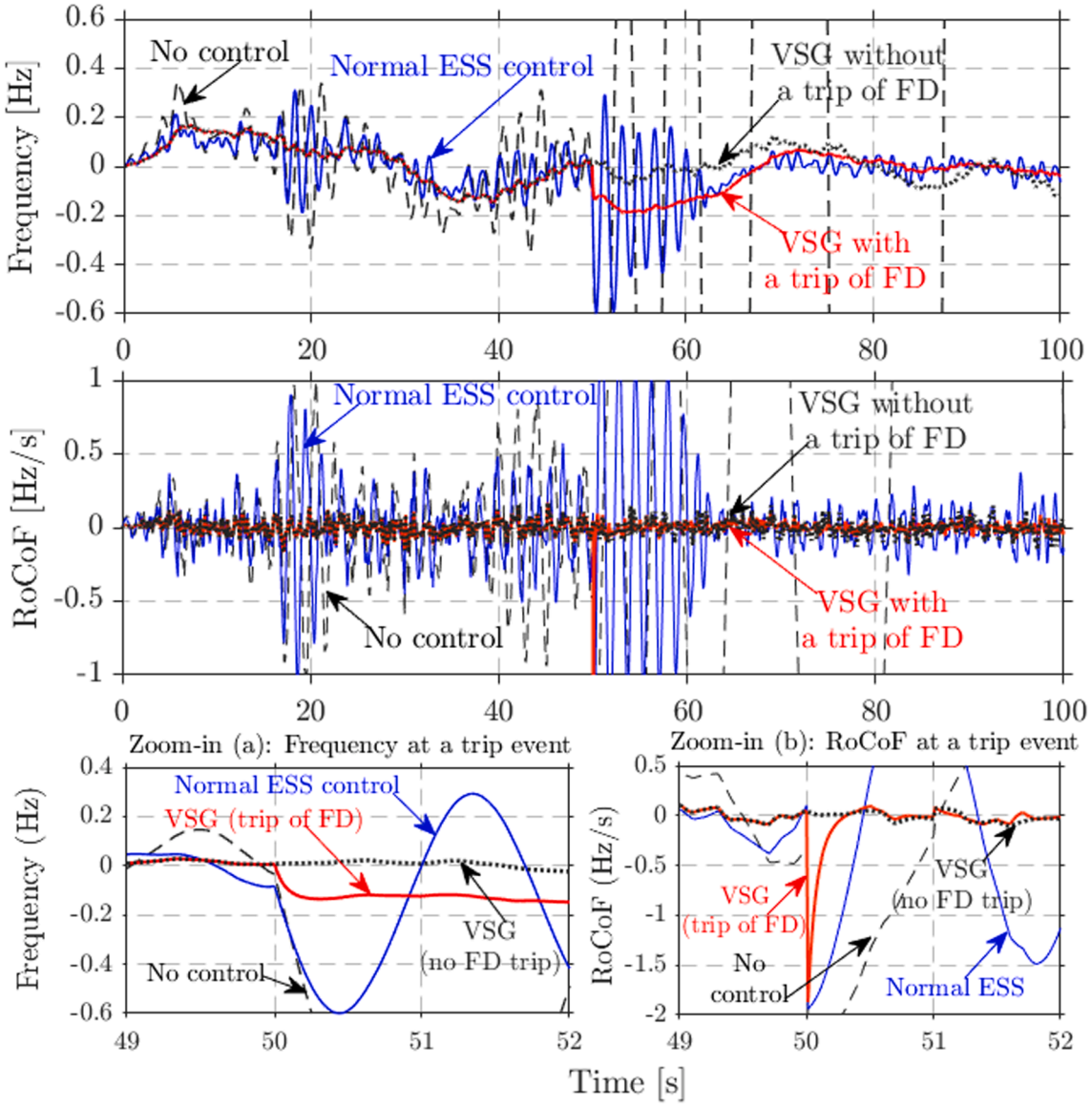


Fig. 12. MG response during the FD-dominated generation trip.

$\Delta\bar{f}(s) = (\Delta f_1 e^{-st_{D,1}} + \Delta f_2 e^{-st_{D,2}} + \Delta f_3 e^{-st_{D,3}} + \Delta f_4 e^{-st_{D,4}})/4$, It is noted that the t_D is the communication delay time [30–32]. In this study, it is assumed that the e^{-st_D} is added to the frequency deviation of each area. The VSG is installed in all control areas and the $\Delta\bar{f}$ is used as an control input to the VSGs. Other system conditions remain the same as the previous case of the interconnected system. To clearly observe the time delay effect, the sudden load change of −0.05p.u. is applied as the disturbance in area 2.

As seen in Fig. 16, when the frequencies of each area/node encounter the delay time (t_D) from 0.5 s and 2.5 s, the mean frequency response of the MG is deteriorated by possessing larger overshoots with oscillations. Similarly, the higher delay time causes a slow response to the VSG unit. In addition, the VSG is more oscillating and consumes higher power to stabilize the MG frequency. During the period of small time delay (i.e., 0.5 s), the MG frequency stability does not significantly degrade, where the emulated VSG power is quite similar to the case of no delay time. This is because the extended VSG control could counteract the disturbances that occurred at the close time. When the time delay escalates over 1.5 s, the mean frequency oscillation with high overshoots could be significantly observed, where a higher VSG power is required. In such a condition, the extended VSG control could not precisely eliminate the forced and other disturbance effects due to the huge time lag, leading to further frequency and power oscillations with high amplitudes.

4.7. VSG sensitivity over maximum rocof, res penetration, inertia reduction under FO effect

In the last case study, the potential advantage of VSG control over FO components is confirmed under wide ranges of RES-based CIG penetration and inertia reduction. Evidently, when the FD is not contained in the MG (see Fig. 17), approximately 60 % of RES-based CIG power penetration in the case of no control and normal ESS control, the MG stability becomes instability with high-frequency deviations over ± 1 Hz. After around 60 % of the RES penetration, the maximum RoCoF over 3 Hz/s occurs in the case of normal ESS control, while the maximum RoCoF is non-linearity and unpredictable in the case of no control. As illustrated in Fig. 18, when the FD occurs in the RES-based CIG, its penetration level is significantly limited to 40 % of the capacity in the case of no control and normal ESS control. Apparently, the normal ESS control can resist the impact of higher RES penetration up to 50 % before reaching severe frequency deviations over ± 2 Hz. After 50 % of the RES penetration, the maximum RoCoF over 4 Hz/s could be observed in the

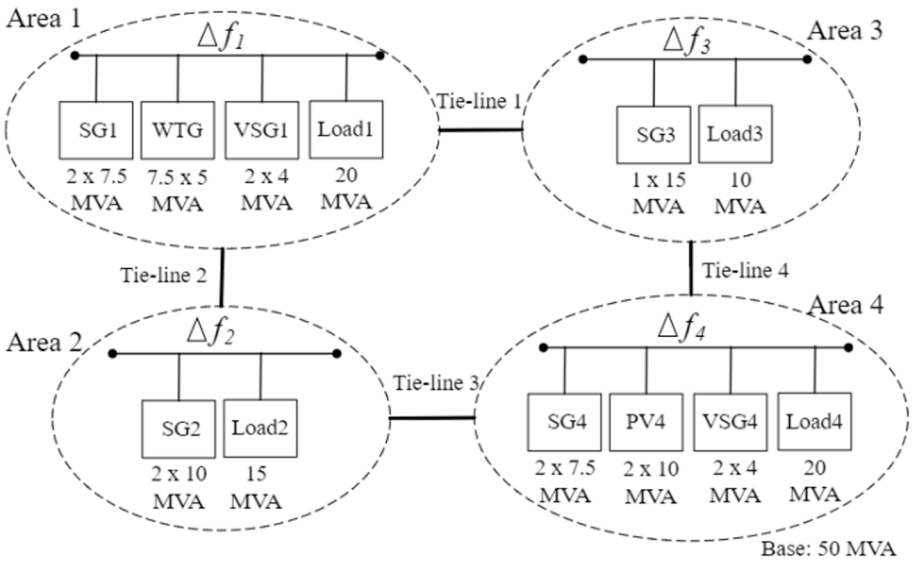


Fig. 13. The studied MG-based four-area interconnected system.

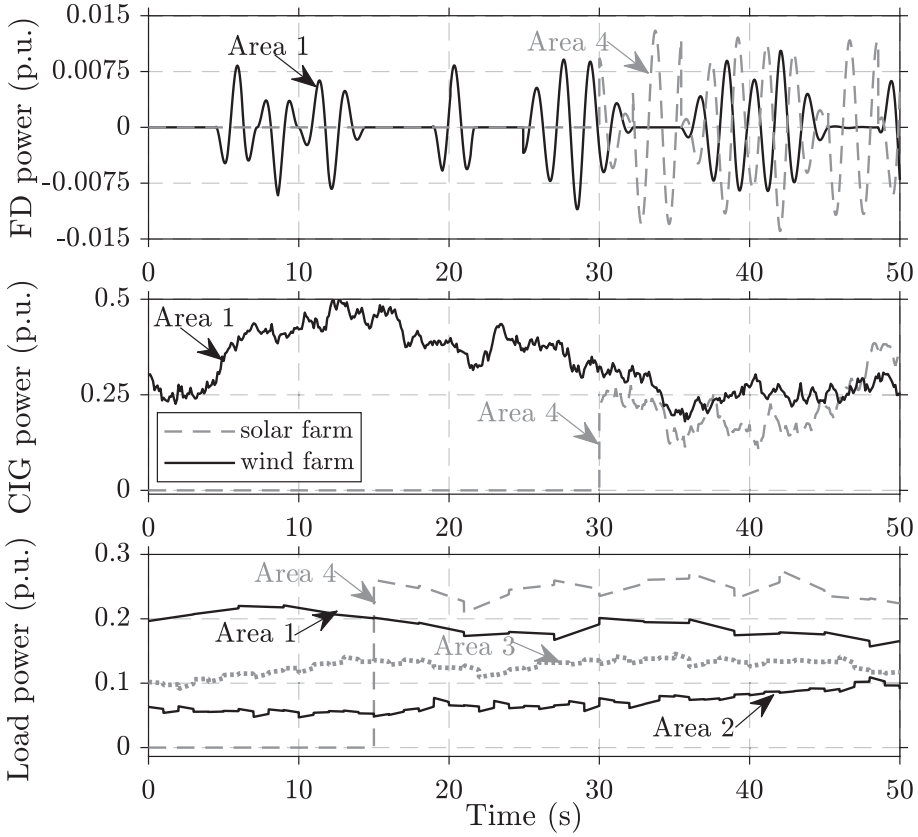


Fig. 14. Power disturbances in the studied four-area interconnected system.

case of normal ESS control, while the maximum RoCoF is non-linearity in the case of no control. Applying the VSG control confirms that the MG stability can preserve the frequency deviation within ± 0.2 Hz with the maximum RoCoF ≤ 1 Hz/s up to 80 % of the RES penetration. Thus, it is validated that the extended VSG method can significantly enhance the damping of Fos, where the FO magnitude/energy is effectively attenuated within a safe level. Subsequently, it could provide the system operators with adequate time to pinpoint the particular FO source, leading to the FD elimination in the low-inertia MG.

5. Conclusions

This research demonstrates the superiority of the extended VSG control to suppress the forced components in the presence of a low-

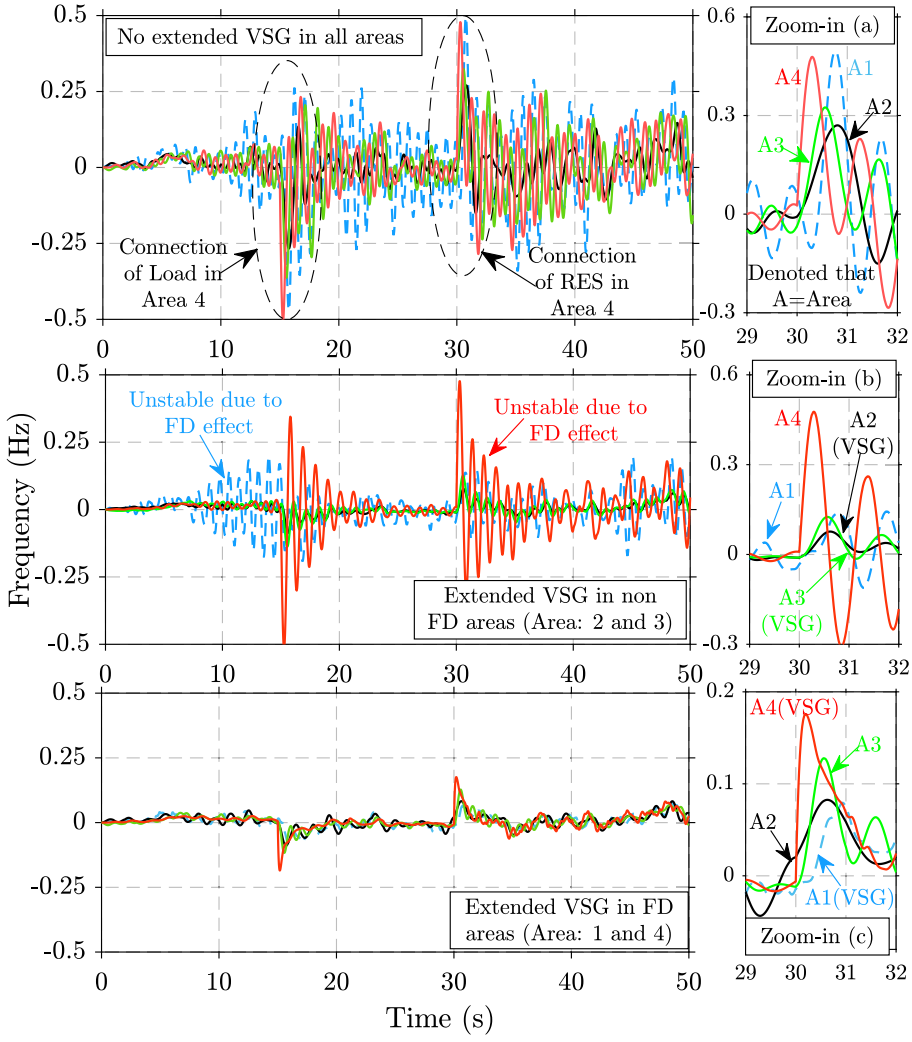


Fig. 15. Frequency responses of four-area MGs with extended VSG control against the forced component and low inertia property.

inertia MG. Comprehensive FO characteristics have been examined in time and frequency domains under the presence of a low-inertia MG with high portions of RES-based CIGs. As a result, the frequency fluctuation and RoCoF frequently exceed acceptable ranges even if forced components produced by CIGs are very low, especially when the total system inertia reduces and power outputs from the CIGs increase. By applying the extended VSG control, the virtual forced component can be emulated to directly eliminate the actual forced component effect, resulting in more robust MG resiliency and performance against the high portion of RES-based CIGs. The main concluding remarks can be listed as follows:

i) The FO effect is periodic, while the severity and behavior of the FO significantly depend on the system inertia and disturbance frequency. The FO can exhibit an identical oscillation response to the characteristics of FDs if the system inertia is close to zero. The FO response is not always influenced by the system's dominant mode during the FO event. Irrespective of the closeness of the dominant frequency, the FO effect on damping is miniature, but there is no significant impact on the mode frequency. Hence, the FO can significantly affect the damping and frequency of the system mode during and after the FO events.

ii) By analyzing the extended VSG control parameters, the virtual forced component could effectively diminish the oscillation effects from FDs, where the increasing J_{VSG} and D_{VSG} result in lower frequency and RoCoF excursions, improving the stability of MG operation and control.

iii) During the low-inertia MG with multiple FD types, the extended VSG control shows superior performance for not only the sake of enhancing MG stability and robustness but also further FO suppression. The MG frequency deviation could be maintained within the acceptable frequency operating standard in standalone and interconnected MGs. Moreover, the extended VSG control demonstrates faster power response than ESS droop control, resulting in lower RoCoF and frequency deviations. Consequently, a smoother power deviation from the SG unit could be achieved, leading to less mechanical stress on the generator and turbine components. Consequently, the stability margin of a low-inertia MG could be significantly augmented.

iv) When the FO-dominated generation trip, the trip/disconnection of the FD source is not necessary in the case of extended VSG control, since the VSG could effectively regulate the MG frequency within the acceptable operating standard. On the other hand, the trip of the FD source can also be applied in the case of

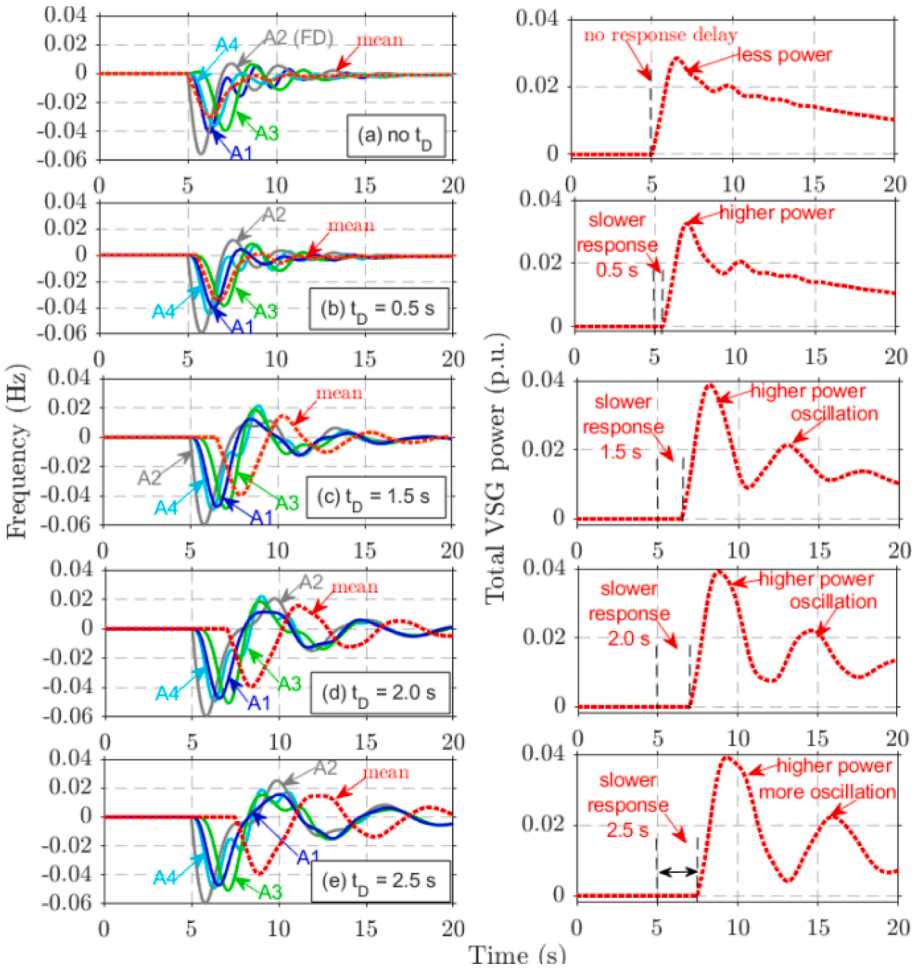


Fig. 16. A control delay effect on the mean frequency deviation over VSG power generation.

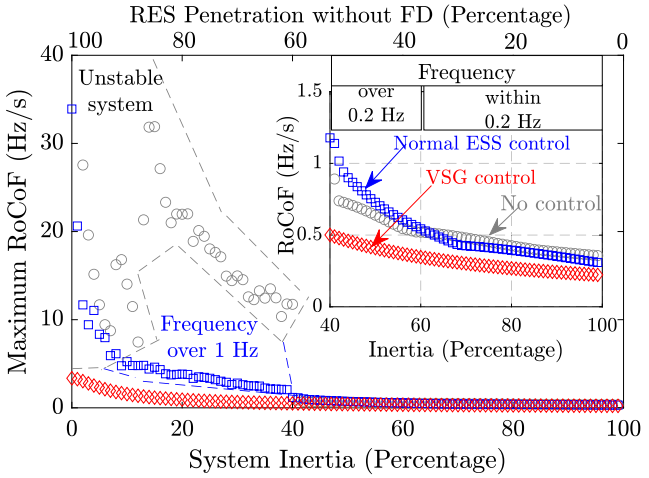


Fig. 17. Maximum RoCoF under the changes of MG inertia and RES penetration without FD effect.

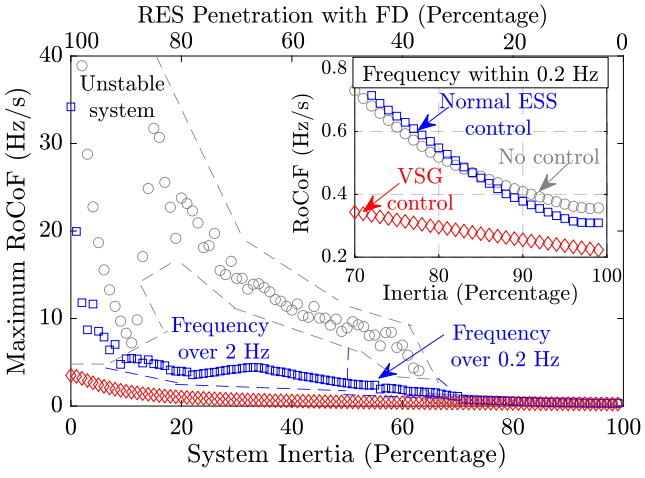


Fig. 18. Maximum RoCoF under the changes of MG inertia and RES penetration with FD effect.

VSG control. This could alleviate the transient effect with a smaller amplitude and setting time than those of no control and ESS droop control.

v) When the FD source cannot be located and eliminated instantly, the extended VSG control can be deployed to modulate the active power of RES-based CIG, where the interminable FO components are effectively arrested by the virtual forced component. Subsequently, the FO energy can be alleviated to a secure level without threatening system stability and requiring further power measurement techniques.

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CRediT authorship contribution statement

Thongchart Kerdphol: Conceptualization, Formal analysis, Investigation, Methodology, Validation, Data curation, Software, Visualization, Writing – original draft. **Issarachai Ngamroo:** Supervision, Funding acquisition, Project administration, Data curation, Conceptualization, Writing – review & editing. **Tossaporn Surinkaew:** Conceptualization, Investigation, Methodology, Software, Validation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix

The linearized state-space model of a microgrid in Section 3 employs the detailed state-space matrices and state variables. To understand the MG dynamic and control behaviors, the detailed state-space matrices corresponding to the MG modelling of (9) and (10) are derived and given as:

$$A = \begin{bmatrix} -\frac{D_{MG}}{2H_{MG}} & \frac{1}{2H_{MG}} & 0 & 0 & \frac{1}{2H_{MG}} & \frac{1}{2H_{MG}} & \frac{1}{2H_{MG}} \\ 0 & -\frac{1}{T_{tb}} & \frac{1}{T_{tb}} & 0 & 0 & 0 & 0 \\ -\frac{1}{R \bullet T_{gov}} & 0 & -\frac{1}{T_{gov}} & \frac{1}{T_{gov}} & 0 & 0 & 0 \\ \frac{1}{\beta \cdot K_{sec}} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{T_{WTG}} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{T_{PV}} \end{bmatrix},$$

$$B^c = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ \frac{1}{T_{CIG}} \\ 0 \\ 0 \end{bmatrix}, B^d = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}, B^e = \begin{bmatrix} 0 & 0 & -\frac{1}{2H_{MG}} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ \frac{1}{T_{WTG}} & 0 & 0 \\ 0 & \frac{1}{T_{PV}} & 0 \end{bmatrix}, C = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

The details of state variables are given as:

$$\Delta x = [\, \Delta\bar{f} \quad \Delta P_{SG} \quad \Delta P_{gov} \quad \Delta P_{sec} \quad \Delta P_{VSG} \quad \Delta P_{WTG} \quad \Delta P_{PV} \quad]^T$$

$$\Delta w = [\, \Delta v \quad \Delta ir \quad \Delta P_{TL} \,]^T$$

where, D_{MG} is the MG load damping constant, H_{MG} is the MG inertia constant, T_{tb} is the time constant of a turbine in the SG unit, R is the droop constant for primary frequency control, T_{gov} is the time constant of a governor in the SG unit, β is the bias factor for secondary frequency control, K_{sec} is the integral gain of secondary frequency control, T_{WTG} is the time constant of the WTG, T_{PV} is the time constant of the PV, ΔP_{SG} is the power deviation from the SG unit, ΔP_{gov} is the change of control signal from the governor, ΔP_{sec} is the change of control signal from secondary control, ΔP_{WTG} is the generated power deviation from the WTG, ΔP_{PV} is the generated power deviation from the PV, Δv is the change of wind velocity, Δir is the change of solar irradiation and ΔP_{TL} is the power deviation from the total load in the MG.

Interested readers can find details and proofs in [30]. More complex models (e.g., non-linear models or higher orders) of MG components can be found in [36–38].

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